

**Text, Web Content and Link Analytics:** Introduction, Text Mining, Web Mining, Web Content and Web Usage Analytics, Page Rank, Structure of Web and Analyzing a Web Graph.  
**TB2:** Ch5: 5.2,5.3, Ch 9: 9.1-9.4

#### PRACTICAL COMPONENT OF IPCC

| Sl.NO | Experiments (Java/Python/R)                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Install Hadoop and Implement the following file management tasks in Hadoop:<br>Adding files and directories<br>Retrieving files<br>Deleting files and directories.<br><b>Hint:</b> A typical Hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities. |
| 2     | Develop a MapReduce program to implement Matrix Multiplication                                                                                                                                                                                                                                                                         |
| 3     | Develop a Map Reduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day.                                                                                                                                                                                                   |
| 4     | Develop a MapReduce program to find the tags associated with each movie by analyzing movie lens data.                                                                                                                                                                                                                                  |
| 5     | Implement Functions: Count – Sort – Limit – Skip – Aggregate using MongoDB                                                                                                                                                                                                                                                             |
| 6     | Develop Pig Latin scripts to sort, group, join, project, and filter the data.                                                                                                                                                                                                                                                          |
| 7     | Use Hive to create, alter, and drop databases, tables, views, functions, and indexes.                                                                                                                                                                                                                                                  |
| 8     | Implement a word count program in Hadoop and Spark.                                                                                                                                                                                                                                                                                    |
| 9     | Use CDH (Cloudera Distribution for Hadoop) and HUE (Hadoop User Interface) to analyze data and generate reports for sample datasets                                                                                                                                                                                                    |

#### Course outcomes (Course Skill Set):

At the end of the course, the student will be able to:

1. Identify and list various Big Data concepts, tools and applications.
2. Develop programs using HADOOP framework.
3. Make use of Hadoop Cluster to deploy Map Reduce jobs, PIG, HIVE and Spark programs.
4. Analyze the given data set and identify deep insights from the data set.
5. Demonstrate Text, Web Content and Link Analytics.

#### Assessment Details (both CIE and SEE)

The weightage of Continuous Internal Evaluation (CIE) is 50% and for Semester End Exam (SEE) is 50%. The minimum passing mark for the CIE is 40% of the maximum marks (20 marks out of 50) and for the SEE minimum passing mark is 35% of the maximum marks (18 out of 50 marks). A student shall be deemed to have satisfied the academic requirements and earned the credits allotted to each subject/course if the student secures a minimum of 40% (40 marks out of 100) in the sum total of the CIE (Continuous Internal Evaluation) and SEE (Semester End Examination) taken together.

#### CIE for the theory component of the IPCC (maximum marks 50)

- IPCC means practical portion integrated with the theory of the course.
- CIE marks for the theory component are **25 marks** and that for the practical component is **25 marks**.
- 25 marks for the theory component are split into **15 marks** for two Internal Assessment Tests (Two Tests, each of 15 Marks with 01-hour duration, are to be conducted) and **10 marks** for other